Aqua Pulse

Smart Aquaculture Monitoring System

1. Introduction

- Aqua Pulse is a smart aquaculture monitoring and control system designed for fish farms, ponds, and lakes.
- It helps farmers and managers monitor water quality in real-time, receive alerts, and manage devices remotely.
- The system combines IoT devices, web platform, and mobile control to improve productivity and reduce risks in aquaculture.

2. How the System Works

The system has two main parts:

A. Monitoring Platform (Software)

- A central system that collects and manages all data
- Users can log in and view information through dashboards
- Data is stored securely in a database
- Alerts are generated automatically when water quality is not safe

B. IoT Field Device (Hardware)

- A smart device installed near or on the pond
- It collects water data and sends it to the server using mobile network (4G)
- Works even in remote areas without Wi-Fi

3. What the System Does

- Monitors water quality in real-time
- Tracks key parameters like:

1. Temperature
2. pH level
3. Turbidity (water clarity)

- Sends data continuously to the backend system
- Generates alerts when values go beyond safe limits
- Sends email notifications to users
- Displays all data in easy-to-use dashboards

4. User Roles

The system supports different types of users:

Manager

- Controls the overall system
- Manages users, devices, and data

Owner

- Views pond data and alerts
- Monitors farm performance

Agent (Field Staff)

- Handles devices in the field
- Monitors assigned ponds

5. Smart Boat Integration

- A smart boat can be controlled using a mobile app
- The boat can:

1. Move on water (forward, backward, left, right)
2. Carry sensors for data collection

- Controlled using MQTT communication
- Built using MIT App Inventor mobile app

6. Key Features

- Role-based login system
- Device registration and assignment
- Pond/site management
- Real-time water monitoring
- Threshold-based alert system
- Email notifications
- Dashboard views for all user roles
- GSM-based remote data transmission
- Mobile app-based boat control

7. Technology Stack

- Backend: FastAPI
- Frontend: React + Vite
- Database: PostgreSQL
- Hardware Controller: ESP32
- Connectivity: 4G GSM Module (A7670C / SIM7670)
- Mobile App: MIT App Inventor
- Communication Protocol: MQTT

8. Future Enhancements

We are planning to expand Aqua Pulse with advanced features to make it more powerful and useful:

Advanced Water Quality Parameters

- Addition of more sensors to monitor:

1. Nitrogen
2. Ammonia
3. Electrical Conductivity (EC)
4. Salinity

- This will provide more accurate and detailed water analysis for better farming decisions

Agent Smart Tablet Device

- Provide agents with a portable tablet device
- The tablet will have:

1. Built-in sensors
2. Data collection capability

- Agents can:

3. Directly collect water data from the field
4. Upload data instantly to the system

- This reduces dependency on fixed devices and improves flexibility

Underwater Smart Boat (Submarine Mode)

- Upgrade the boat into a submarine-type device
- Capabilities include:

1. Monitoring fish/prawn movement at deeper levels
2. Observing health conditions underwater
3. Capturing real-time underwater data
4. Performing deep water quality testing

- Helps in better analysis of aquatic life behavior and environment

9. Conclusion

Aqua Pulse is a complete smart solution for modern aquaculture.

It helps in:

- Improving water quality monitoring
- Reducing risks in farming
- Increasing productivity and efficiency

With future enhancements like advanced sensors, smart agent devices, and underwater monitoring, Aqua Pulse aims to become a next-generation intelligent aquaculture platform.